Deep Neural Network Approach in Electrical Impedance Tomography-based Real-time Soft Tactile Sensor

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IROS MACAU 2019

KAIST

1. Introduction

2. Method

3. Result

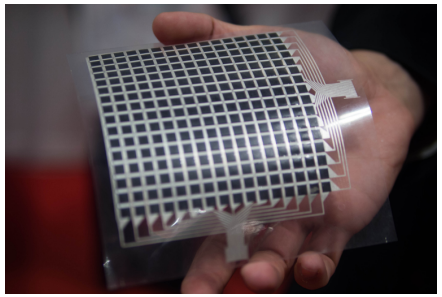
4. Conclusion

Sensor for whole-body tactile sensing

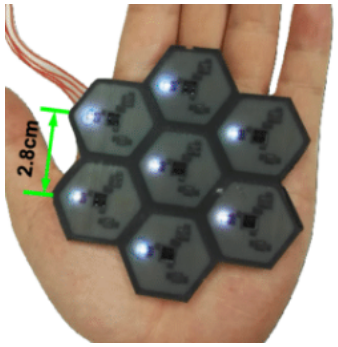
Electrical Impedance Tomography (EIT)

- medical imaging with **only boundary measurements**
- monitoring conductivity difference of piezoresistive materials
- **high durability, easy manufacturability** and **large size with fewer electrodes**

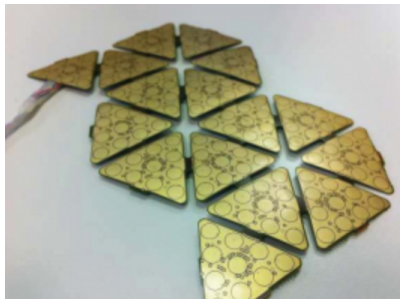
Conventional



© PST Sensors 2017

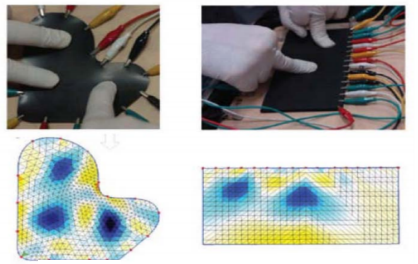


Mittendorfer, 2013

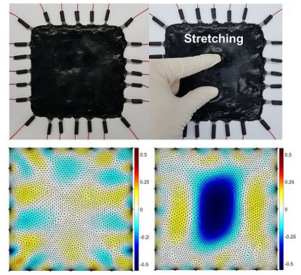


Metta, 2008

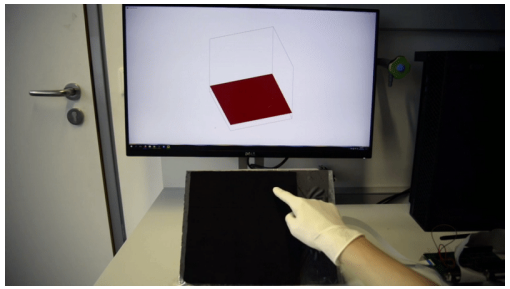
EIT-based



Alirezaei 2007

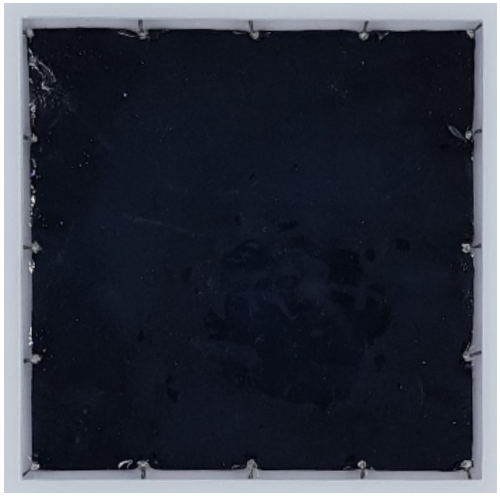


Oh & Yang, 2018

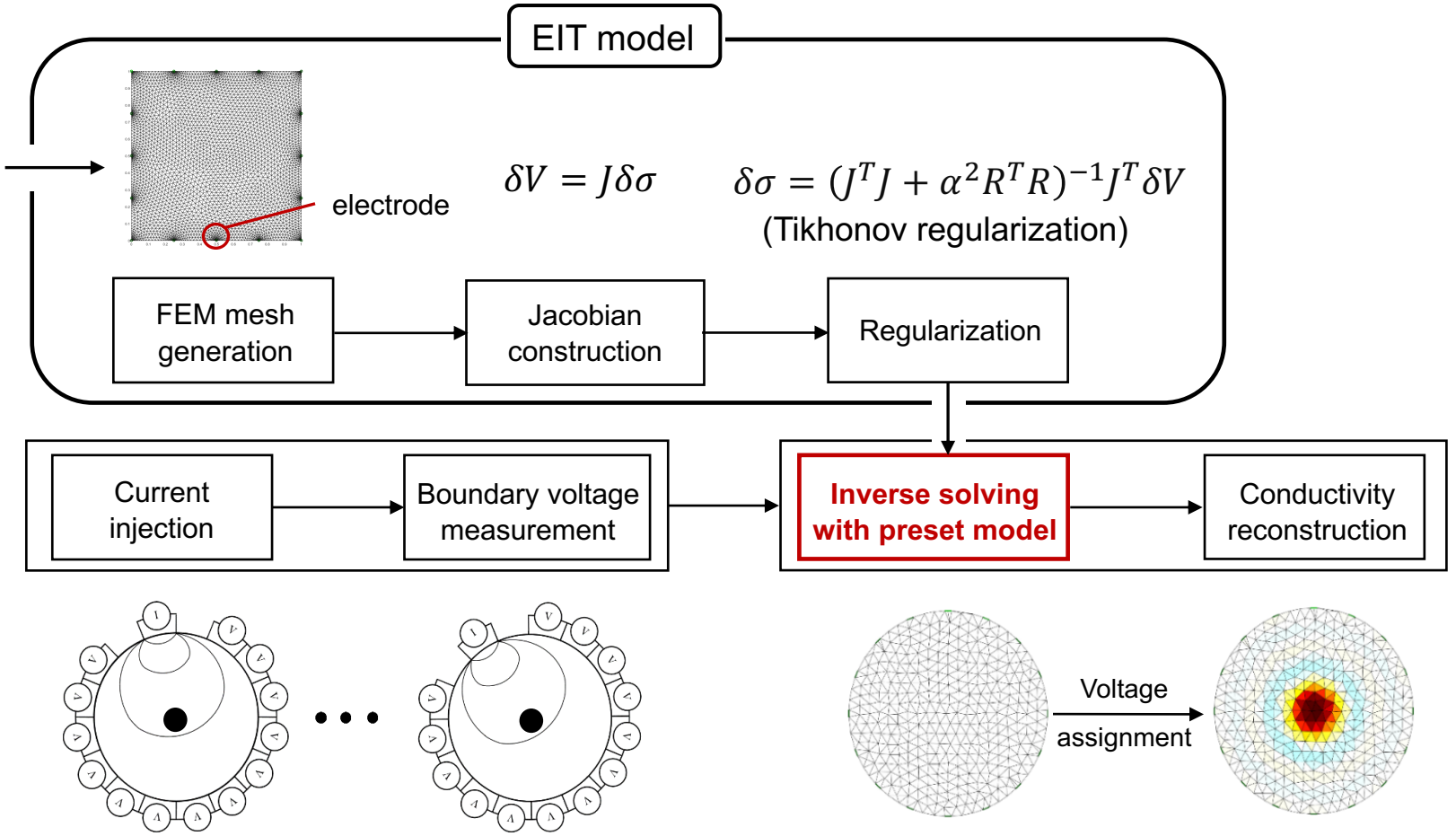


Lee & Park, 2018

Operating principle



Sensor hardware

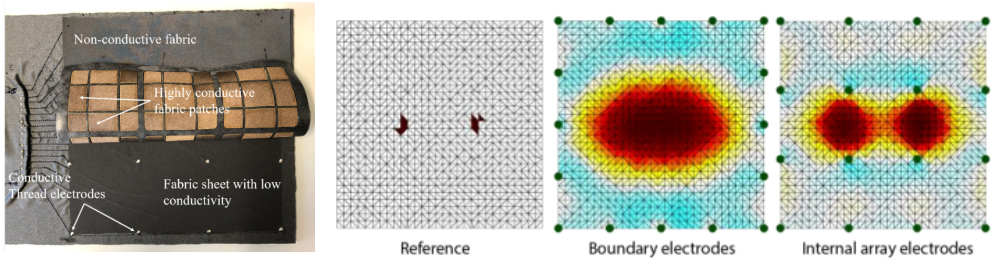


Drawback : Low measurement performance

Low spatial resolution and high noise sensitivity due to ill-posed nature of EIT inverse problem

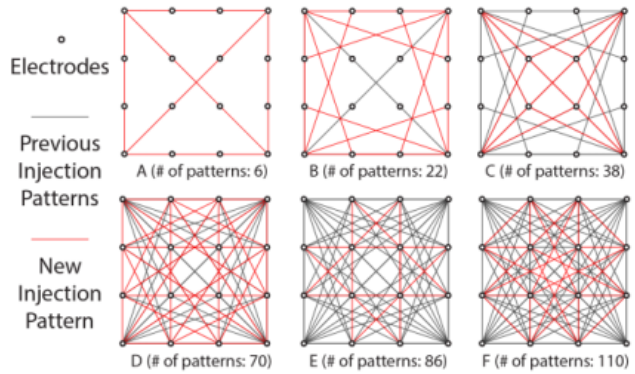
Hardware perspective

- Insertion of internal electrodes [Lee & Park, 2018] [Tawil, 2011]



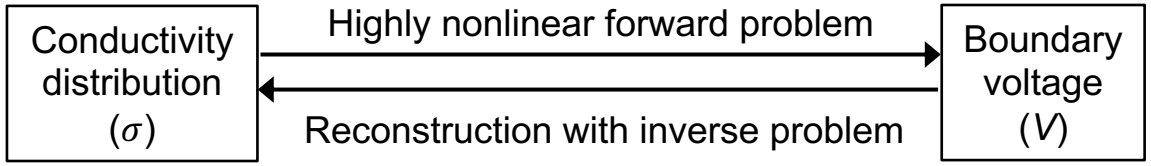
- Optimal stimulation / measurement pattern

[Lee & Park, 2018]
[Gisser, 1987]



Data processing perspective

- Post-processing the output image [Martin, 2017]
- Nonlinear EIT sensing model

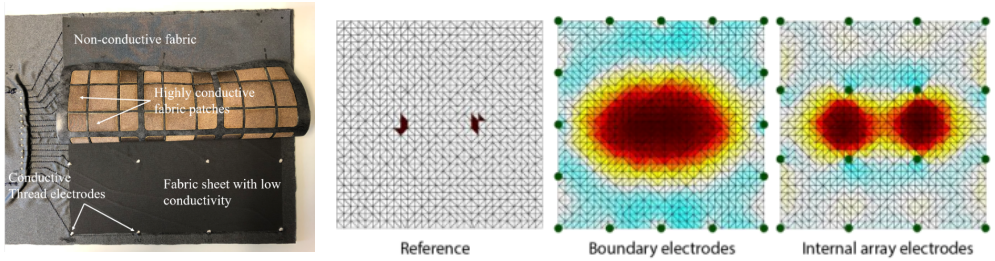


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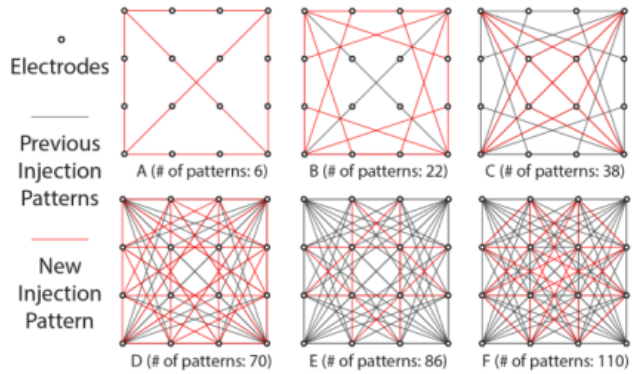
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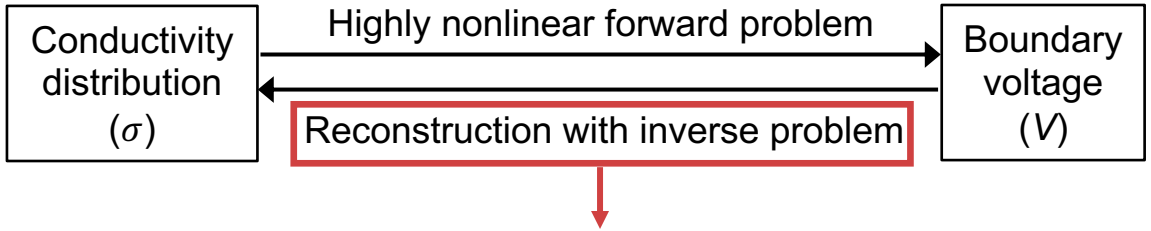
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Data processing perspective

- Post-processing the output image [Martin, 2017]
- Nonlinear EIT sensing model



[Conventional approach : Linearized assumption σ =inv(J)*V
nonlinear approach : PDIPM
Particle Swarm Optimization [Martins, 2012]
Simulated annealing [Martin, 2015]
⋮

Long computational time,
not suitable for tactile sensing applications

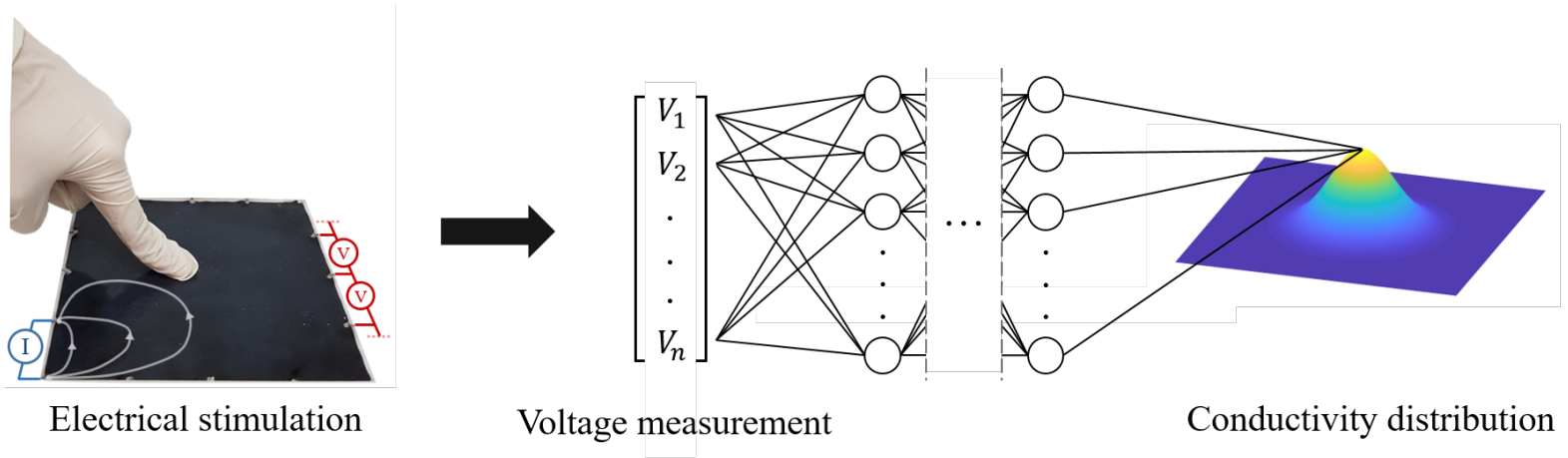
Research goals

Use of Deep Neural Network (DNN) approach for fast and more accurate sensing performance

- Sim-to-real approach with FEM simulation of the sensor
- Improving sensing performance in spatial resolution, sensitivity, and generalized reconstruction

Implementation of the proposed sensing model into real-time sensing application

- FPGA-based fast sensor operation with DNN sensing model



1. Introduction

2. Method

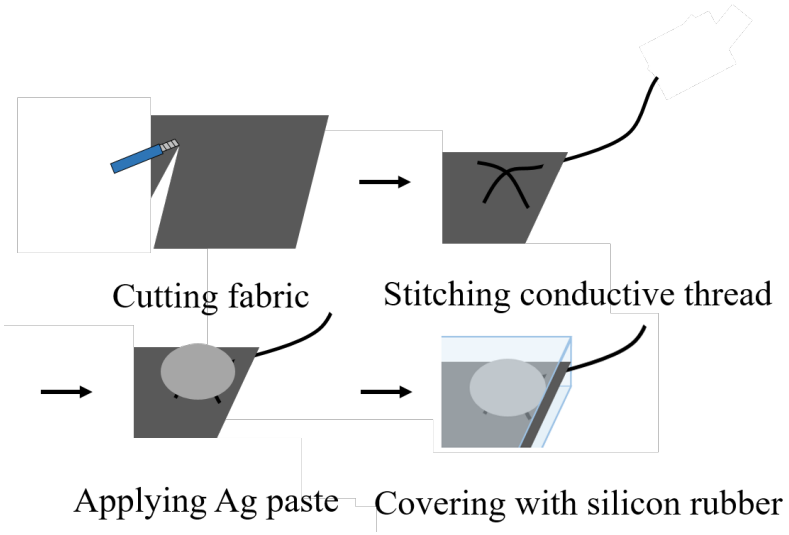
3. Result

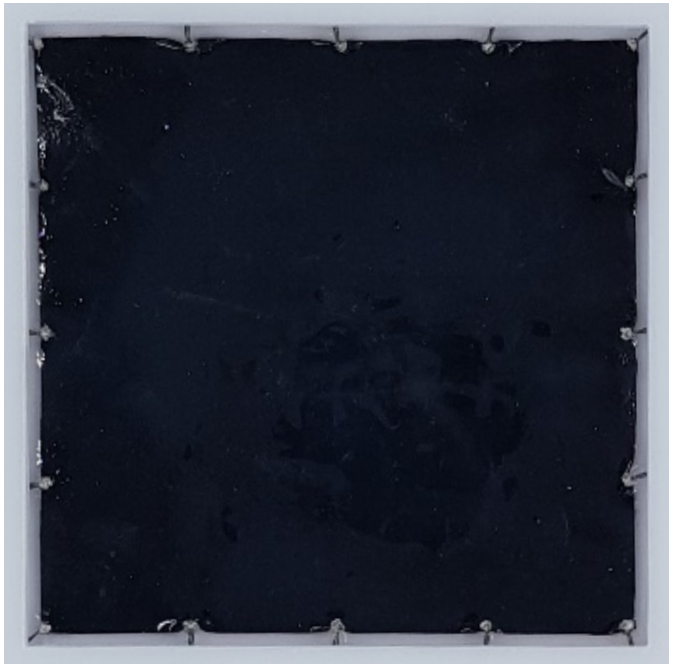
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Sensor fabrication

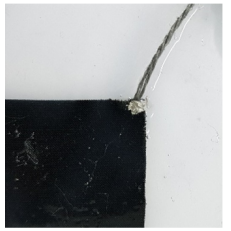
Simple procedure with attachment of electrodes to conductive domain

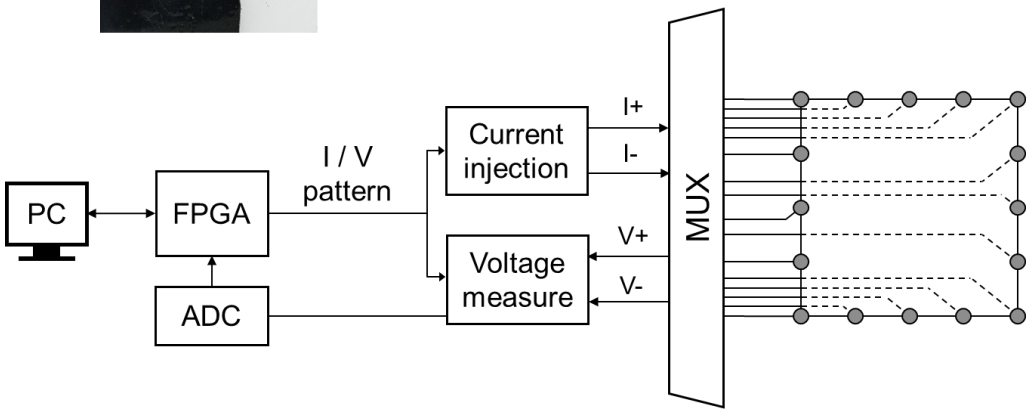
- Commercial **piezoresistive fabric** as conductive domain (COM-14112, Eeonyx, USA)
- Stitching **conductive thread** for robust connection to fabric ,Covering the domain with **silicon rubber**





size : 14cm x 14 cm



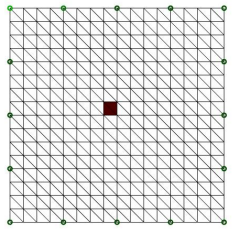


Dataset generation

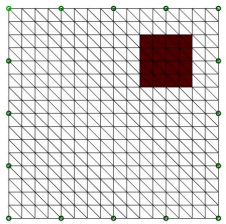
Forward solving with FEM

- Utilization of **preset shape of distribution for efficient dataset generation**
- Simulation conducted with EIDORS opensource, MATLAB [Vauhkonen, 2001] [Adler, 2006]
- Total 76,776 dataset that maps **256 conductivity elements-to-208 voltage measurements**

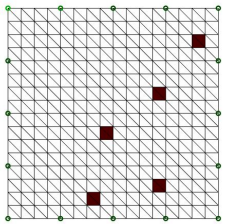
Conductivity perturbation($\delta\sigma$)



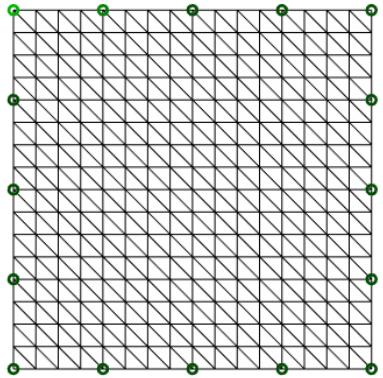
Single elements



Square distribution

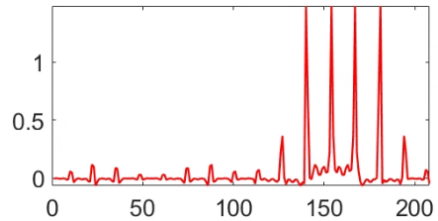


Multi elements selection

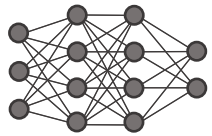


FEM simulation with Maxwell's equation

Boundary voltage(δV)



DNN training

δV  $\delta\sigma$

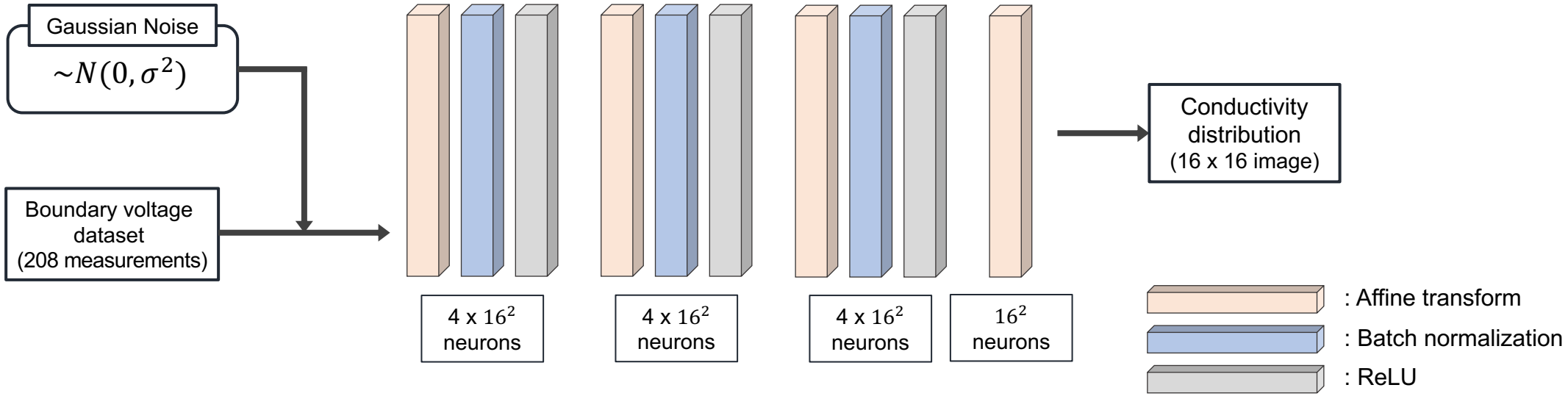
Training

Neural network architecture

- Fully connected layer with ReLU activation function
- Batch normalization for fine convergence, high generalization capability [Ioffe, 2015]

Training info.

- Adam optimizer, learning rate of 0.01
- mini-batch training,size of 100 (Low size for high generalization capability) [Kesker, 2017]
- Keras backend Tensorflow with single NVIDIA GeForce GTX 1080ti



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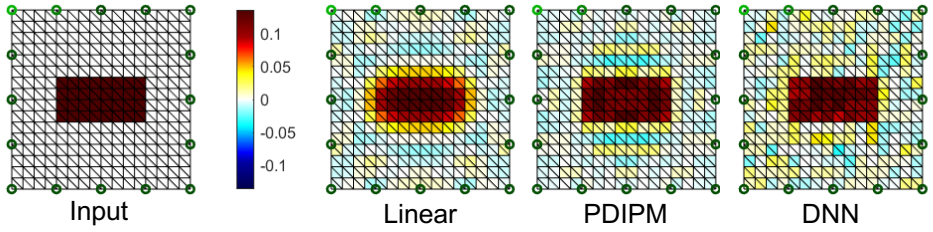
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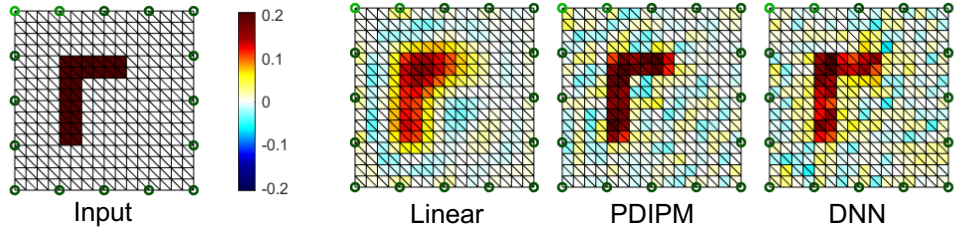
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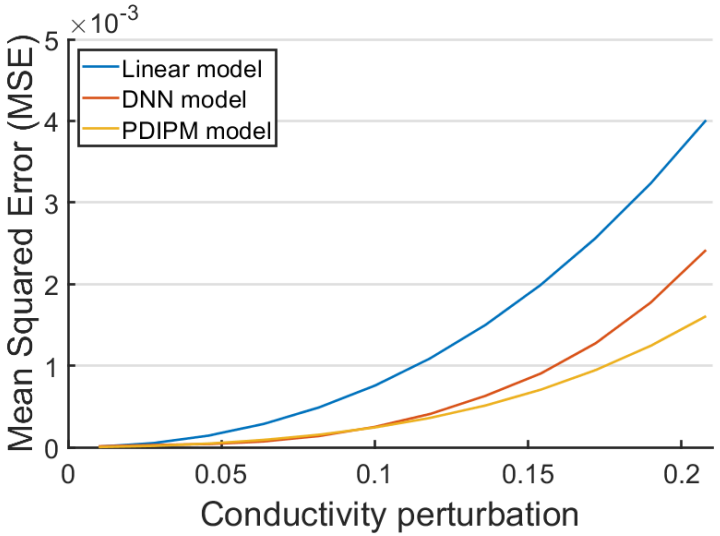
Reconstruction accuracy

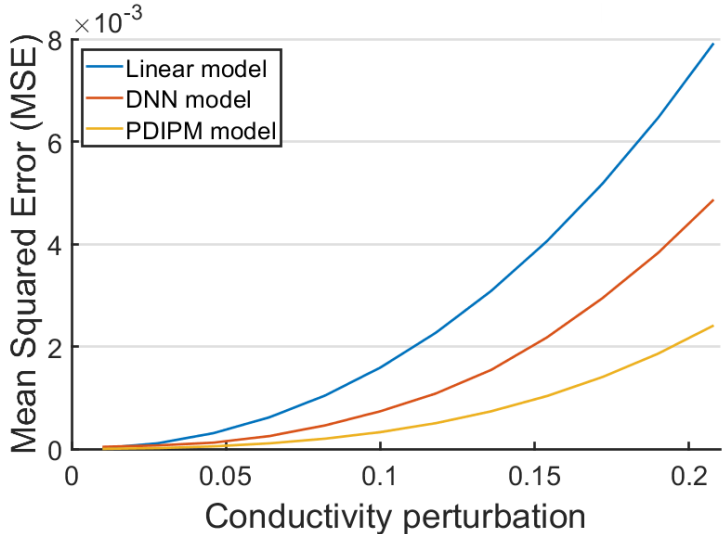
Simulation study

- Alternatives to validate the performance (lack of instruments for measuring conductivity distribution)
- Mean squared error (MSE) between estimated image and reference image
- **Removed regularization effect** for fair comparison without heuristic decision





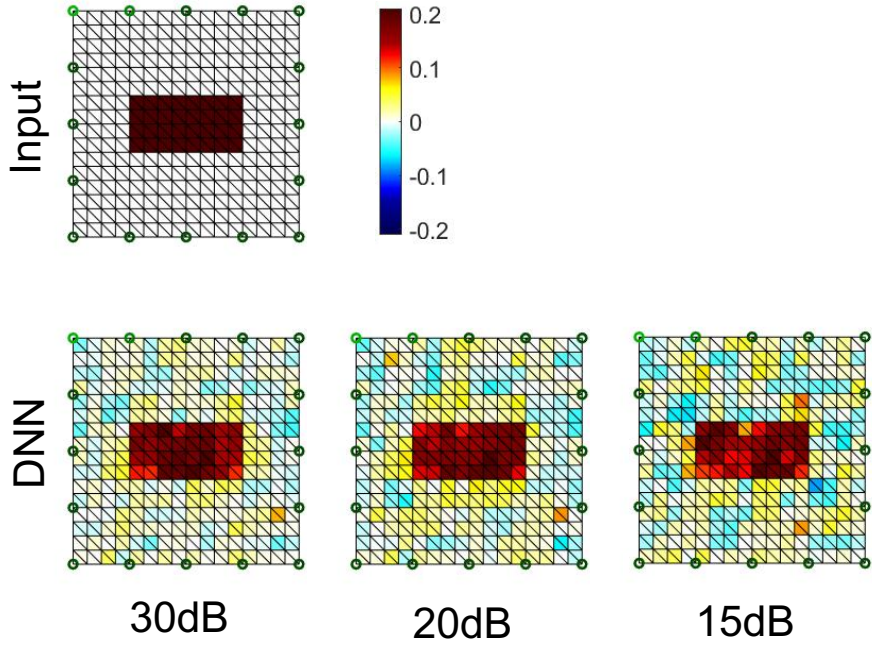


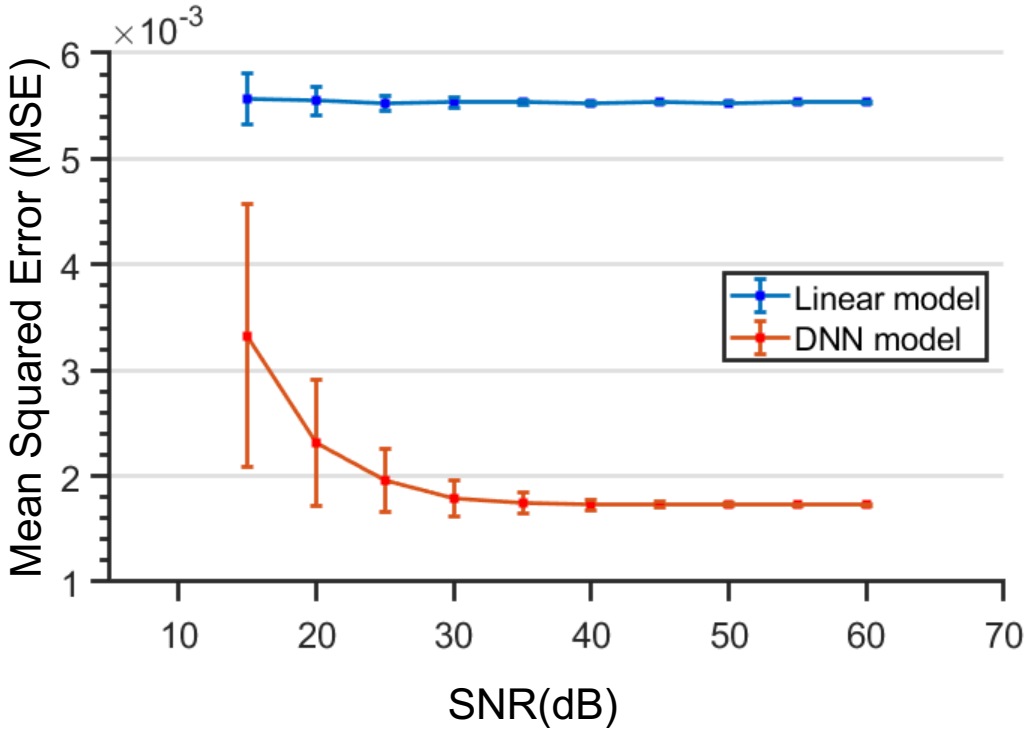


Noise robustness

Simulation study

- Performance validation with standard deviation in 100 numbers of repetitive reconstruction
- Higher noise sensitivity, but fine reconstruction observed
- The image error is uniformly pervasive, because of **the lack of the information of smoothness**





Real-time operation

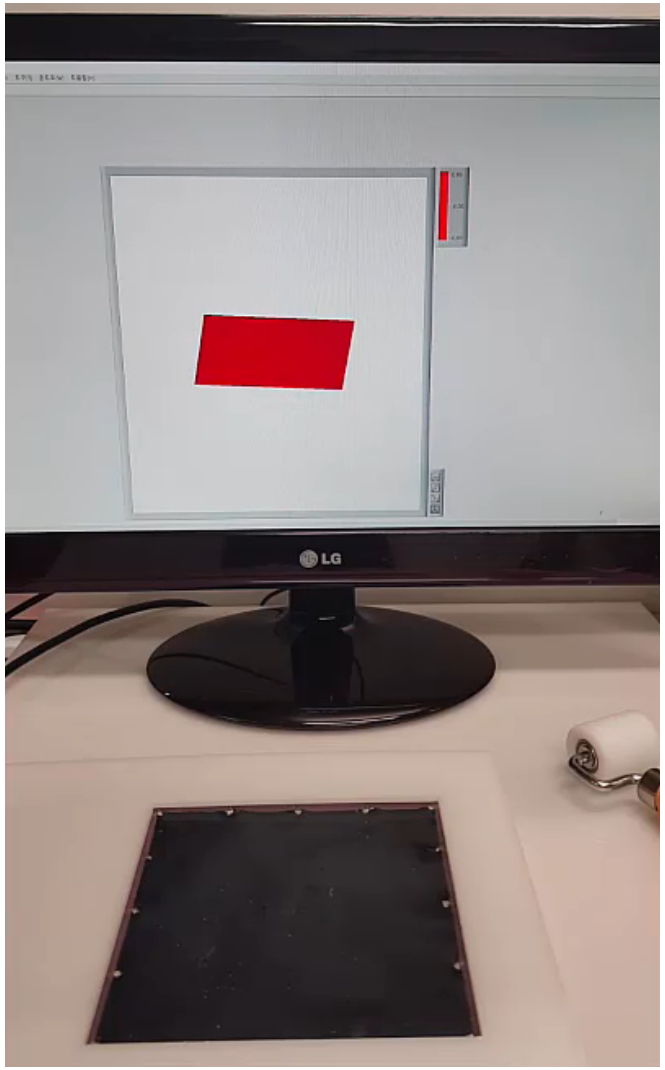
Computational time comparison

- CPU computational time of nonlinear methods
- Potentially high frame rate capability observed in DNN model

	Computation time(s)
Iterative Gauss-Newton	4.782
PDIPM	0.227
ANN(RBFNN)	0.027
DNN, proposed	0.002

Real-time operation platform

- FPGA-aided fast sensor voltage measurement **(50kHz)**
- Model operation and visualization with Labview platform **(50Hz)**
- Image post-processing with Gaussian spatial filter



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Conclusion

Deep Neural Network-based approach in EIT sensing method

- Higher sensing performance validated with simulation study
- Fast operation, suitable for real-time operation
- Realization of FPGA-based operation platform

Further works

- Use of sensor with lower offset drift and having more uniform base conductivity
- Performance validation in a **real sensor measurement (indentation test, two-point discrimination)**
- Higher framerate by using FPGA acceleration to inference of EIT-NN

EIT-NN : Generalized reconstruction framework

Improvement in DNN model construction

- Fully connected layer **+ Convolution layer**, inspired from AUTOMAP architecture [Zhu, 2018]
- Increase the size of image, 24 x 24
- Novel loss function for **smoothing the image, imposing the well-posedness**
- Stable and robust convergence, more insensitive to initial parameters

Sensor domain transform

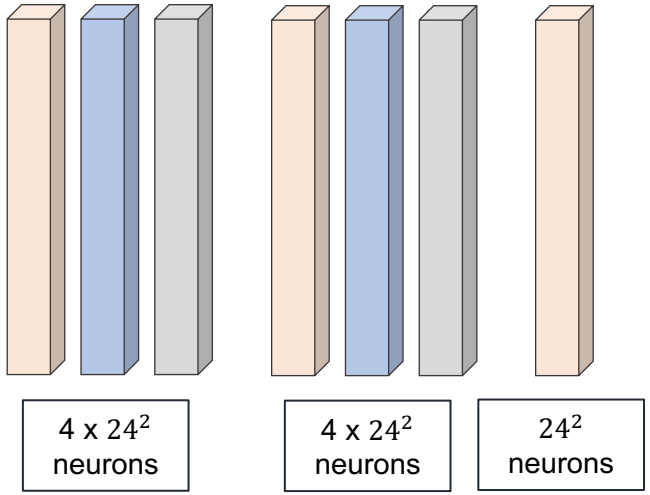
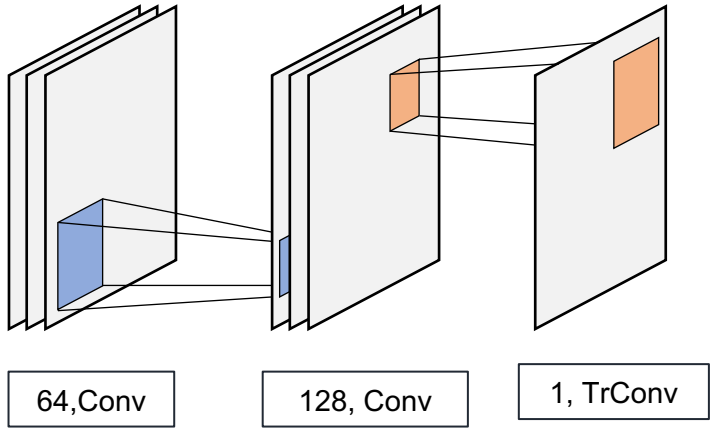
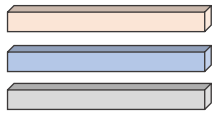


Image processing

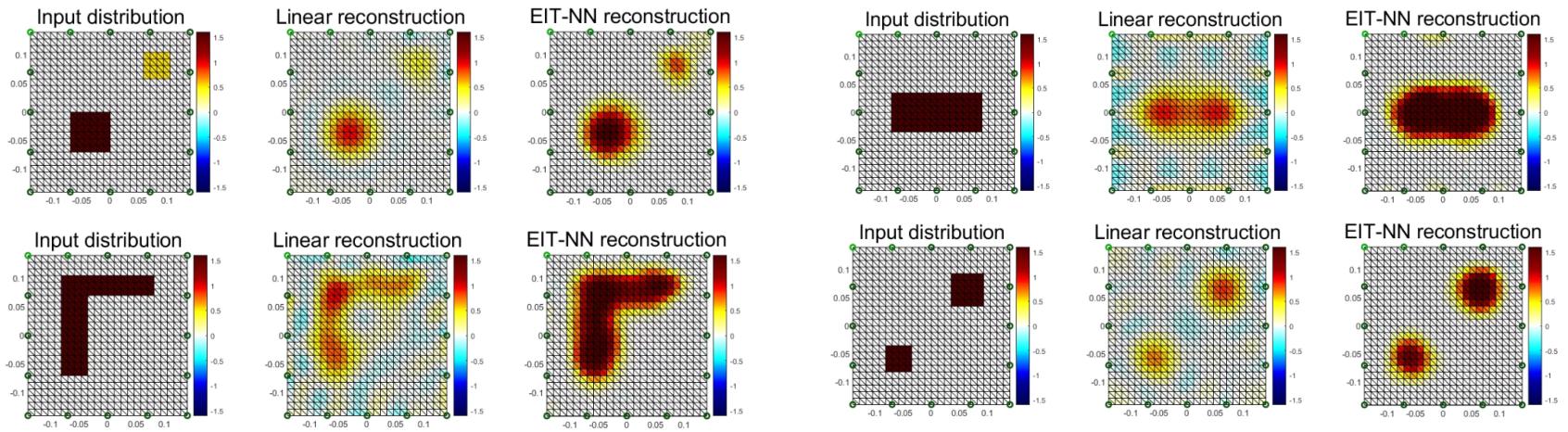


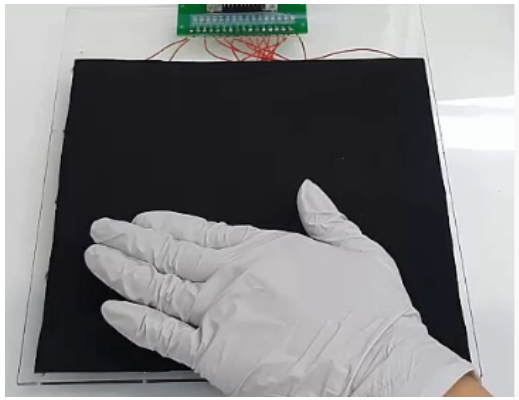


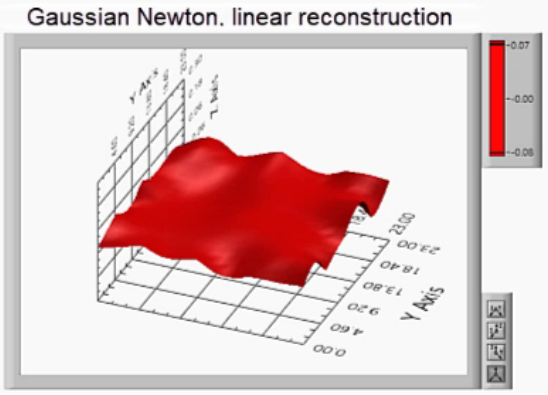
Sensing performance

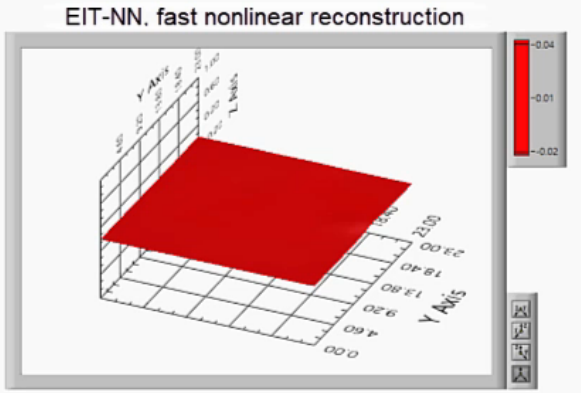
Overall improvement in spatial resolution

- More accurate reconstruction in various scenarios
- Validation in real sensor measurement







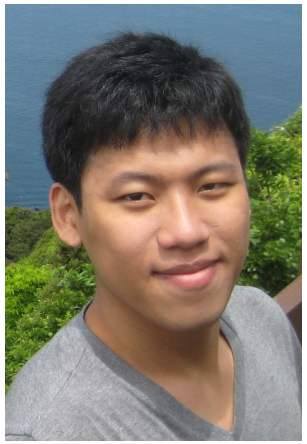


Thank you for your attention



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